

Linking Biosignal Responses to Annotated Traffic Scenario Events in Simulated Autonomous Vehicle Rides

Xiaoxi Wang, Faculty Mentor: Dr. Junfeng Zhao
FURI Research Proposal

1. Research Question and Hypotheses

Autonomous vehicle (AV) safety research has advanced significantly from an engineering perspective, with scenario-based evaluation frameworks providing quantifiable safety metrics across dynamic driving conditions (Wishart et al., 2023, 2025). However, a critical gap remains between system-level safety assessment and passengers' subjective experience of safety. When a real hazard exists, passengers may fail to perceive it; when the system is operating nominally, passengers may still feel unsafe. Understanding when and why perceived safety diverges from objective system performance requires connecting two streams of evidence that AV research has largely kept separate: the specific features of traffic scenario events, and the physiological responses those events produce in passengers.

This study asks: which traffic scenario event features (e.g. TTC time-to-collision, acceleration/deceleration profile, and lane change maneuver) are associated with measurable changes in passenger biosignal responses (e.g. EDA and eye-tracking), and do those changes correspond to perceived safety ratings? Results will serve as the foundational dataset for a larger project led by Dr. Junfeng Zhao and involving a group of research students.

Three hypotheses guide data collection and analysis:

H1: Phasic skin conductance response (SCR) amplitude and frequency will increase significantly in the post-onset window of high-intensity hazard events (cut-ins with low TTC) relative to baseline highway cruise segments.

H2: Gaze scanning breadth will increase during novel AV maneuvers (e.g., AV-initiated lane changes and sudden deceleration events) compared to baseline cruise segments, reflecting heightened monitoring in response to unfamiliar or abrupt vehicle behavior.

H3: Participants reporting lower continuous perceived safety ratings will show higher tonic skin conductance level (SCL) and broader gaze scanning in the preceding event window compared to participants reporting higher perceived safety under matched scenario conditions.

2. Literature Review

Engineering frameworks for AV safety assessment have advanced significantly, with scenario-based metric systems providing quantifiable performance scores across dynamic driving conditions (Wishart et al., 2023, 2025). These methods establish the system-safety side of the human-machine equation. Endsley (2017) argues, increasing automation does not remove humans from the loop, it repositions them as monitors whose perceptual and physiological state remains a safety-relevant variable. On the human side, He et al. (2022) demonstrated that perceived risk and trust in automated driving can be modeled using behavioral and physiological measures, and Deng et al. (2024) showed that trust states in human-automation interaction are detectable through multimodal signal streams including voice and behavioral responses.

Despite this foundation, existing studies examine human risk perception and system safety largely in isolation: engineering evaluations generate scenario-level safety metrics without incorporating passenger physiological response, while human factors studies measure perceived safety without anchoring those measures to specific annotated driving events and their objective features. This study addresses that gap by treating event-biosignal-perception alignment as its primary contribution.

3. Research Plan and Methods

The SURI study contributes the foundational data infrastructure: a synchronized, event-annotated biosignal dataset collected in the ASU Polytechnic driving simulator, which will feed directly into the broader project's biosignal-based perceived safety prediction pipeline. The existing experimental devices including the driving simulator, Pupil Labs Neon eye tracker, and Shimmer Galvanic skin response (GSR) are shown in Figure 1.



Figure 1 Data Collection Equipment: Eye Tracker, Driving Simulator, GSR Device

Platform and participants. Participants (target 8-10) ride as passengers and do not control the vehicle. A pre-session questionnaire assesses prior AV experience as a between-subjects moderator. Biosignal devices. A GSR device records EDA. The Pupil Labs Neon eye tracker captures binocular gaze, yielding road fixation percentage, gaze scanning breadth, and pupil diameter.

Scenario design. Scenario types span the event feature space including highway cruise, cut-in, with various TTC, hard braking, sudden acceleration, automatic lane change.

Perceived safety measures. A continuous thumb-operated slider provides real-time perceived safety ratings aligned to the biosignal timeline. Post-scenario surveys capture retrospective holistic judgments and serve as ground truth for model validation.

Event annotation and alignment. Events are identified from simulator state logs and annotated by scenario class, TTC, acceleration/deceleration profile, and maneuver type. Mixed-effects regression tests whether event features predict biosignal response and perceived safety ratings, with participant as a random effect.

Expected deliverables. Deliverables include: 1) a synchronized, event-annotated biosignal dataset from participants formatted for the broader project's pipeline; 2) a reusable event annotation protocol; 3) preliminary regression results testing H1–H3. 4) a device setup and synchronization protocol document for subsequent data collection on the lab's instrumented AV.

4. Fulton Research Theme

This study falls under the “Data” research theme. The core contribution is a multimodal data collection and alignment pipeline: GSR and eye-tracking signals are synchronized to simulator event markers, annotated against scenario features, and analyzed using event-locked time-series methods and mixed-effects regression. The raw sensor streams will be transformed into a structured dataset that reveals how human perceptual states respond to an autonomous system.

References

- Wishart, J., Rahimi, S., Swaminathan, S., **Zhao, J.**, Frantz, M., Singh, S., & Como, S. G. (2025). Dynamic driving task assessment scores for scenarios navigated by an OEM ADS-equipped vehicle. *SAE Technical Paper*.
- Wishart, J., **Zhao, J.**, Woodward, B., O'Malley, G., Guo, H., Rahimi, S., & Swaminathan, S. (2023). A proposed safety case framework for automated vehicles with pillars of safety management system, design methods, and scenario-based testing. *IEEE International Automated Vehicle Validation Conference*.
- He, X., et al. (2022). Modeling perceived risk and trust in automated driving using behavioral and physiological measures. *Transportation Research Part F: Traffic Psychology and Behaviour*.
- Deng, Z., et al. (2024). Voice-based detection of trust in human–automation interaction. *International Journal of Human-Computer Studies*.
- Endsley, M. R. (2017). From here to autonomy: Lessons learned from human–automation research. *Human Factors*, 59(1), 5–27.

Timeline

Week 1–3: Literature review and equipment learning

Week 4–7: Experiment design and pilot study

Week 8–12: Data collection

Week 11–14: Data analysis

Week 15: Final report

Budget

Experimental equipment and materials: \$250

Participant compensation: \$80

Other materials: \$70

Total: \$400

Personal Statement

I am currently pursuing a degree in Human Systems Engineering at Arizona State University, where I have developed a strong interest in autonomous systems and human factors. I believe that technology and human experience should be considered together. No matter how advanced a system is, it is ultimately designed for human and understanding how users perceive and respond to it is essential.

Transportation affects almost everyone, and autonomous vehicles have the potential to change how people travel. However, technical performance alone does not guarantee user acceptance. Even when a system operates safely, passengers may still feel uncertain if they do not understand its behavior. This has led me to focus on the gap between objective system safety and subjective human safety perception.

During my studies, I have not limited myself to theoretical knowledge. In addition to learning psychology, cognition, and human-centered design, I have emphasized hands-on experience. I believe that understanding user experience requires not only theory, but also real data and effective measurement. Because of this, I am particularly interested in research methods that capture human responses in a direct and reliable way.

This research project is especially meaningful to me because it focuses on structured data collection and analysis. I am interested in learning how to collect data in simulated autonomous vehicle environments and how to use tools such as eye-tracking and skin conductance to capture physiological and behavioral responses. I hope to understand how these biosignals can be aligned with specific traffic events and to explore whether there are observable relationships between event features and human responses.

For me, the most important part of this project is learning how to design experiments, collect reliable data, and perform basic data analysis. I want to build a clear understanding of how different types of data can be connected and interpreted. I believe these foundational skills are essential for studying human behavior in complex systems.

Through this opportunity, I hope to strengthen my skills in experimental design, data collection, and analysis, while gaining hands-on experience with the simulator and biosignal devices. More importantly, I see this project as a starting point for understanding how human responses can be systematically studied in autonomous systems.

In the future, I hope to continue developing in areas related to autonomous driving, safety perception, and human interaction with automated systems. I am interested in research that connects engineering systems with human-centered perspectives. I believe human responses are not only important for evaluation, but can also provide useful insights for improving how systems are designed. I hope this experience will build a strong foundation for my future work in human factors and intelligent systems.

XIAOXI WANG

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SUMMARY

Human Systems Engineering undergraduate at Arizona State University with experience in Human Factors, human-machine interaction, and prototype-driven problem solving. Interested in user trust, safety perception, system feedback, and multimodal measures in safety-critical systems.

EDUCATION

B.S. Human Systems Engineering Arizona State University	Expected Graduation: May 2027 GPA 3.69/4.0
Relevant Coursework: Human-Computer Interaction; Human Systems Integration; Perceptual Systems	
Savannah College of Art and Design - Architecture (Transferred)	2024/09–2025/05
Courses: Design Fundamentals; Architectural Graphics; Form, Space, Order	GPA 4.0/4.0

SKILLS & CERTIFICATIONS

Coding/Web: Python, HTML, CSS, JavaScript Research Equipment: Pupil Labs Neon, Shimmer GSR+/ECG
Tools: Figma, Adobe Photoshop, After Effects, Premiere Pro, AutoCAD, SketchUp, Fusion 360, Revit, Rhino
Statistics & Analysis: EDA, Hypothesis Testing, Effect Size, Power Analysis, Correlation, Regression, ANOVA
Behavioral Foundations: Human Behavior, Perception, Cognition, Sensory Processing, Nervous System
Certifications: PSPO I 2025.12; Adobe Certified Professional Photoshop 2025.05 Valid through 2028.05

EXPERIENCE

Autonomous Systems Lab, ASU: Student Research Assistant	2025/12–Present
- Support autonomous-driving passenger-experience research focused on the relationship between engineering-defined safety and human safety perception. - Conduct literature review on trust, perceived risk, system feedback, and multimodal measures in automated driving. - Learn and prepare research equipment, including Pupil Labs Neon and Shimmer GSR+/ECG, for upcoming experiment setup and data collection.	
GSEICE: User Experience Inter	2025/05–2025/08
- Conducted usability testing and NSF-compliant UX audits for commercial hardware, redesigning manuals to resolve 18% of user pain points and projected a 15% reduction in support queries. - Executed competitive research across Home Depot/Walmart, analyzing 200+ reviews and 5 product lines via Affinity Diagrams to deliver a 15-page market entry and localization strategy report. - Represented company at 4 international trade shows (Chicago/Atlanta), securing 2 high-value B2B orders.	

PROJECTS

Robotic String Instrument Design	2026/01–Present
- Built a five-string robotic instrument using Arduino, servo control, mechanical linkage, Fusion 360, 3D printing, laser cutting, and wood fabrication. - Designed a mechanical hand for string plucking and used flex-sensor input with Arduino control logic to enable remote, gesture-driven performance.	
AI Vision Assist Smart Glasses	2025/10–2025.11
Built Raspberry Pi + Python smart glasses with real-time camera, AI recognition, and voice narration, integrated hardware and usability for visually impaired users.	
AI Conversational Task Manager	2025/05–Present
- Built an AI task-management web app using Next.js, TypeScript, Tailwind CSS, and Supabase. - Enables users to create interactive task lists, update progress, and support later analysis within the same interface through conversation with AI.	

Arizona State University
Unofficial Transcript

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Name: Xiaoxi Wang
Student ID: 1234864061

Print Date: 04/08/2026

Academic Standing: Good Standing
Term Honor: Dean's List

Transfer Credits				
Transfer Credit from SUNY Genesee Community College				
		Attempted	Earned	Points
Course Trans	0.000	Transfer Totals:	0.000	7.000
GPA:				0.000

Transfer Credit from University of North Dakota				
		Attempted	Earned	Points
Course Trans	0.000	Transfer Totals:	0.000	4.000
GPA:				0.000

Transfer Credit from Savannah College of Art & Design				
		Attempted	Earned	Points
Course Trans	0.000	Transfer Totals:	0.000	3.330
GPA:				0.000

Transfer Credit from Keiser University - All Locations				
		Attempted	Earned	Points
Course Trans	0.000	Transfer Totals:	0.000	6.000
GPA:				0.000

Transfer Credit from Community College of Baltimore County				
		Attempted	Earned	Points
Course Trans	0.000	Transfer Totals:	0.000	3.000
GPA:				0.000

Transfer Credit from Tianjin Chengjian University				
		Attempted	Earned	Points
Course Trans	0.000	Transfer Totals:	0.000	31.560
GPA:				0.000

Beginning of Undergraduate Record

2026 Spring

Course	Description	Attempted	Earned	Grade	Points
AME 494	Special Topics	3.000	0.000	NR	0.000
Course Topic:	Expressive Robotics				
HSE 223	Applied Biology Human Behavior	3.000	0.000	NR	0.000
HSE 225	Human Systems Integration	3.000	0.000	NR	0.000
HSE 230	Stats Human Systems Rsrch I	3.000	0.000	NR	0.000
HSE 323	Perceptual Systems	3.000	0.000	NR	0.000
HSE 325	Human-Computer Interaction	3.000	0.000	NR	0.000
LAP 494	Special Topics	1.000	0.000	NR	0.000
Course Topic:	3D Tech for Designers II				
Term GPA:		0.00	Term Totals	0.000	0.000
Cum GPA:		3.69	Cum Totals	18.000	72.890
					66.332

2026 Fall

Course	Description	Attempted	Earned	Grade	Points
HSE 224	Applied Social Science	3.000	0.000	NR	0.000
HSE 290	Exp Method Human Systems Rsrch	3.000	0.000	NR	0.000
HSE 324	Applied Cognitive Science	3.000	0.000	NR	0.000
HSE 424	Human Automation Interaction	3.000	0.000	NR	0.000
HSE 429	Product Design and Evaluation	3.000	0.000	NR	0.000
TMC 498	Pro-Seminar	3.000	0.000	NR	0.000
Term GPA:		0.00	Term Totals	0.000	0.000
Cum GPA:		3.69	Cum Totals	18.000	72.890
					66.332

END OF TRANSCRIPT

2025 Fall					
Course	Description	Attempted	Earned	Grade	Points
BIO 230	How Nature Works	4.000	4.000	B+	13.332
BMV 304	The Practice of Biomimicry	3.000	3.000	B+	9.999
COM 259	Comm Bus & the Professions	3.000	3.000	A-	11.001
FSE 100	Introduction to Engineering	2.000	2.000	A	8.000
HSE 101	Human Systems Engineering	3.000	3.000	A	12.000
IFT 101	Info Tech Programming Logic	3.000	3.000	A	12.000
Term GPA:		3.69	Term Totals	18.000	66.332
Cum GPA:		3.69	Cum Totals	18.000	62.890
					66.332